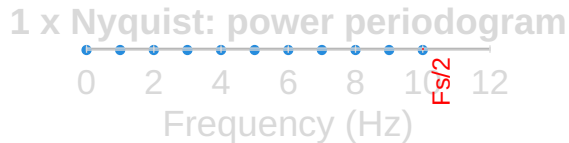
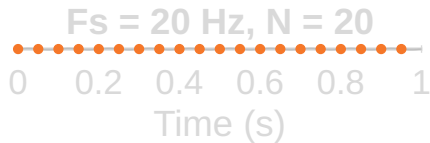
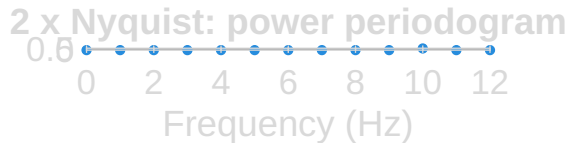


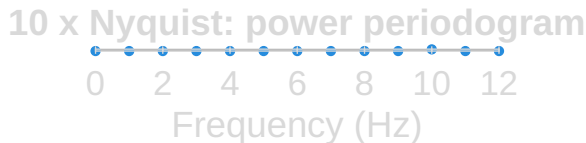
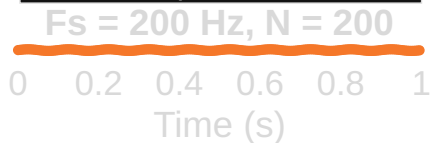
Sinusoid: common amplitude coefficient $A = 1$



— Continuous reference
—● Samples



— Continuous reference
—● Samples



— Continuous reference
—● Samples

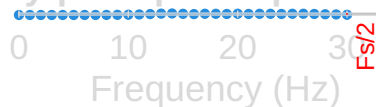
Linear chirp: common amplitude coefficient $A = 1$

$F_s = 60 \text{ Hz}$, $N = 60$



— Continuous reference
—• Samples

1 x Nyquist: power periodogram



$F_s = 120 \text{ Hz}$, $N = 120$



— Continuous reference
—• Samples

2 x Nyquist: power periodogram

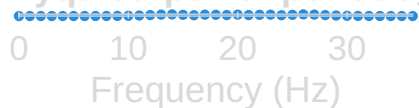


$F_s = 600 \text{ Hz}$, $N = 600$



— Continuous reference
—• Samples

10 x Nyquist: power periodogram



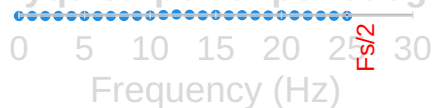
Quadratic chirp: common amplitude coefficient $A = 1$

$F_s = 50 \text{ Hz}$, $N = 50$



— Continuous reference
—● Samples

1 x Nyquist: power periodogram

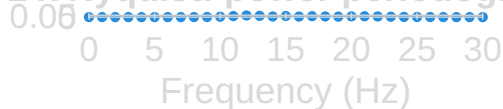


$F_s = 100 \text{ Hz}$, $N = 100$



— Continuous reference
—● Samples

2 x Nyquist: power periodogram

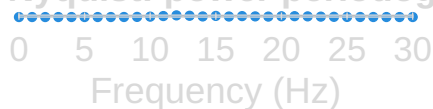


$F_s = 500 \text{ Hz}$, $N = 500$



— Continuous reference
—● Samples

10 x Nyquist: power periodogram

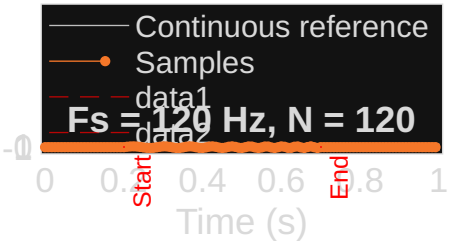


Transient linear chirp: common amplitude coefficient $A = 1$

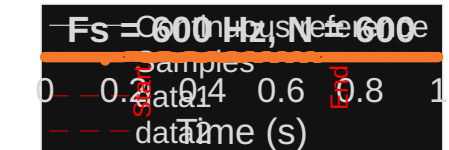
$F_s = 60 \text{ Hz}$, $N = 60$



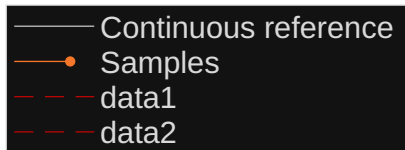
1 x Nyquist: power periodogram



2 x Nyquist: power periodogram



10 x Nyquist: power periodogram



Sine-Gaussian: common amplitude coefficient $A = 1$

$F_s = 60 \text{ Hz}$, $N = 60$



— Continuous reference
—● Samples

1 x Nyquist: power periodogram



$F_s = 120 \text{ Hz}$, $N = 120$



— Continuous reference
—● Samples

2 x Nyquist: power periodogram

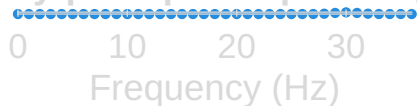


$F_s = 600 \text{ Hz}$, $N = 600$



— Continuous reference
—● Samples

10 x Nyquist: power periodogram



AM: common amplitude coefficient $A = 1$

$F_s = 80 \text{ Hz}$, $N = 80$



— Continuous reference
—• Samples

$F_s = 160 \text{ Hz}$, $N = 160$



— Continuous reference
—• Samples

$F_s = 800 \text{ Hz}$, $N = 800$



— Continuous reference
—• Samples

1 x Nyquist: power periodogram



2 x Nyquist: power periodogram



10 x Nyquist: power periodogram



FM: common amplitude coefficient $A = 1$

$F_s = 120 \text{ Hz}$, $N = 120$



— Continuous reference
—• Samples

$F_s = 240 \text{ Hz}$, $N = 240$



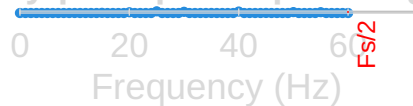
— Continuous reference
—• Samples

$F_s = 1200 \text{ Hz}$, $N = 1200$

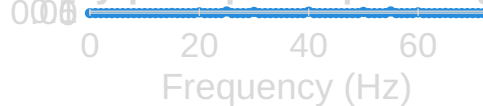


— Continuous reference
—• Samples

1 x Nyquist: power periodogram



2 x Nyquist: power periodogram



10 x Nyquist: power periodogram



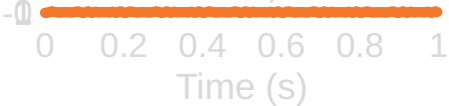
AM-FM: common amplitude coefficient $A = 1$

$F_s = 120 \text{ Hz}$, $N = 120$



— Continuous reference
—• Samples

$F_s = 240 \text{ Hz}$, $N = 240$



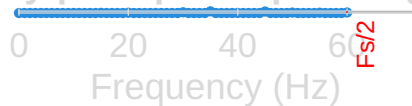
— Continuous reference
—• Samples

$F_s = 1200 \text{ Hz}$, $N = 1200$

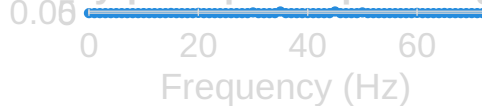


— Continuous reference
—• Samples

1 x Nyquist: power periodogram



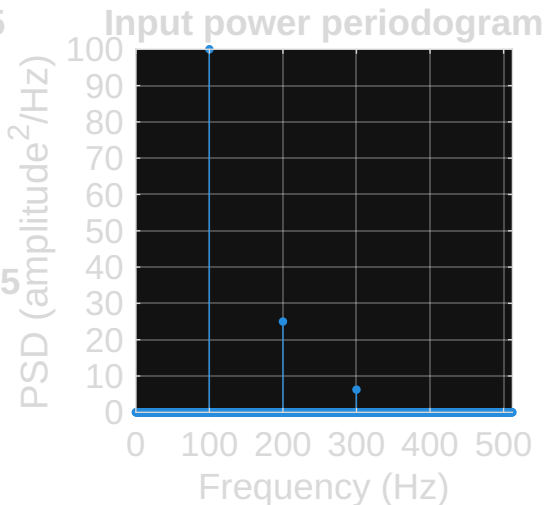
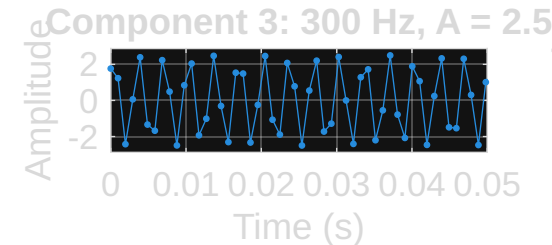
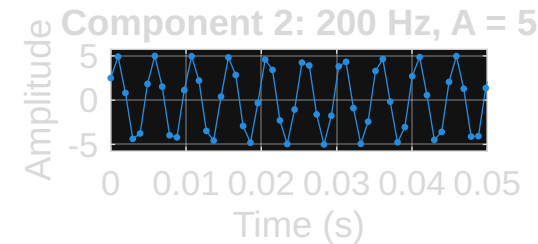
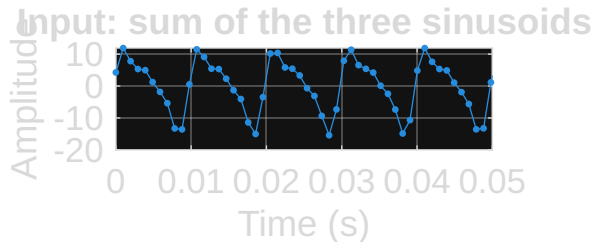
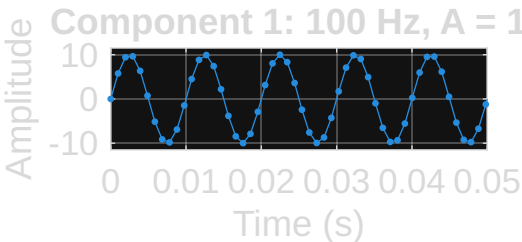
2 x Nyquist: power periodogram

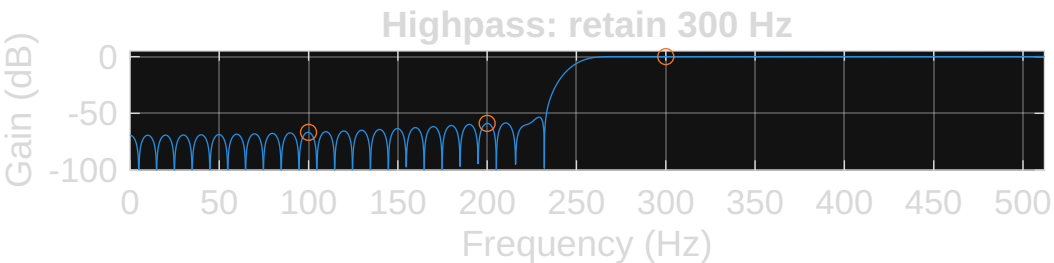
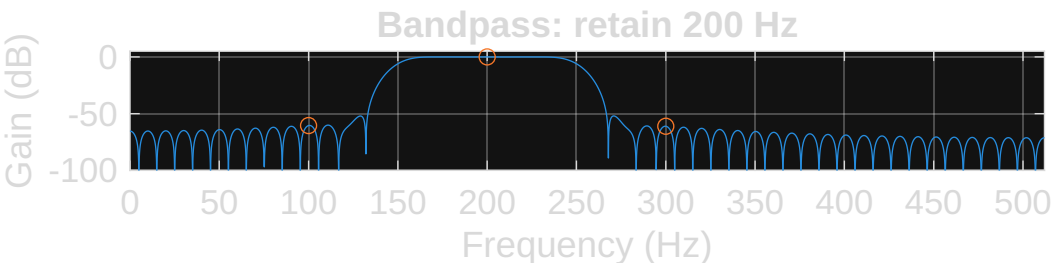
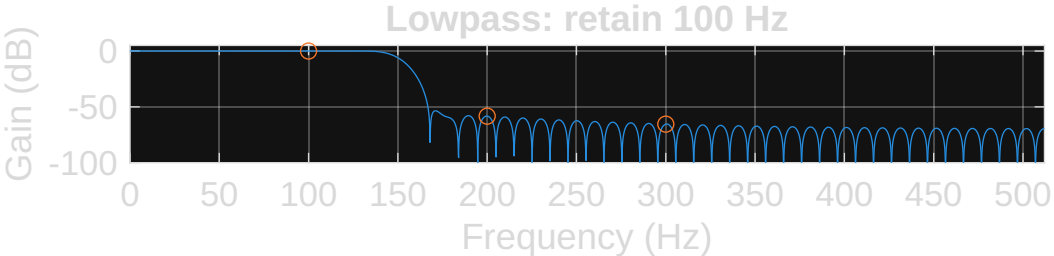


10 x Nyquist: power periodogram



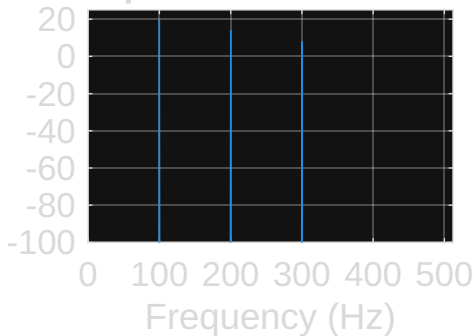
Filtering stage 1: verify the input before designing filters



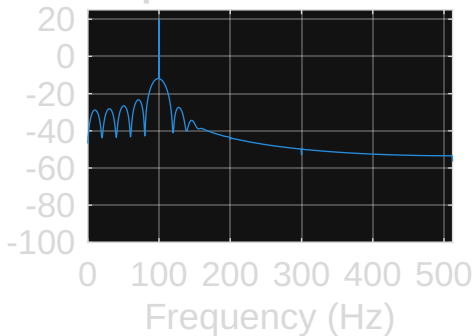


Full-record periodogram: startup transient included

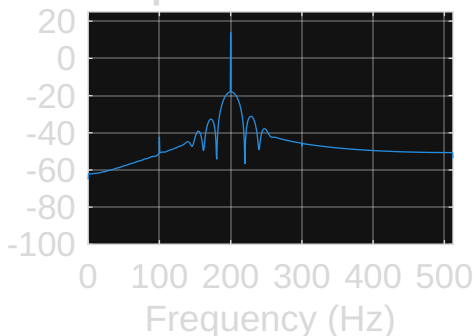
Input: all three tones



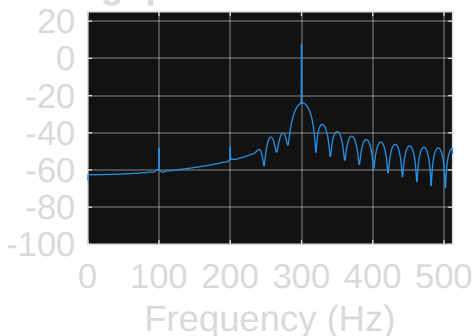
Lowpass: retain 100 Hz

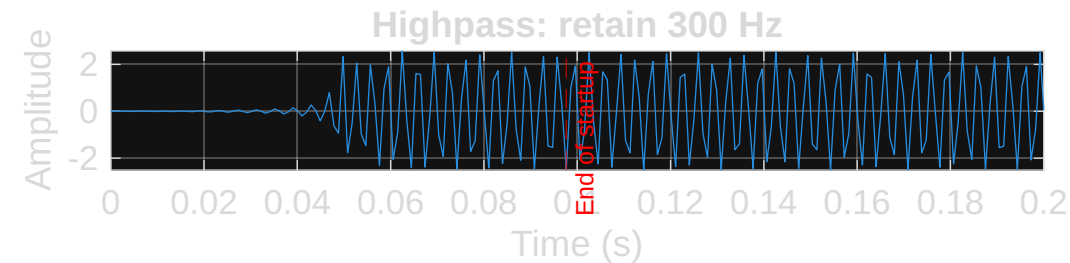
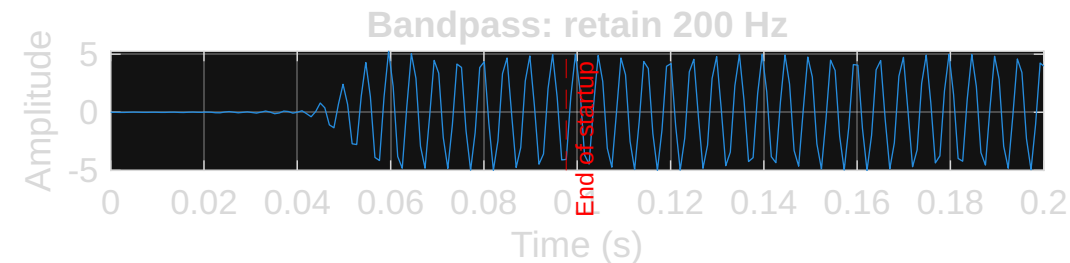
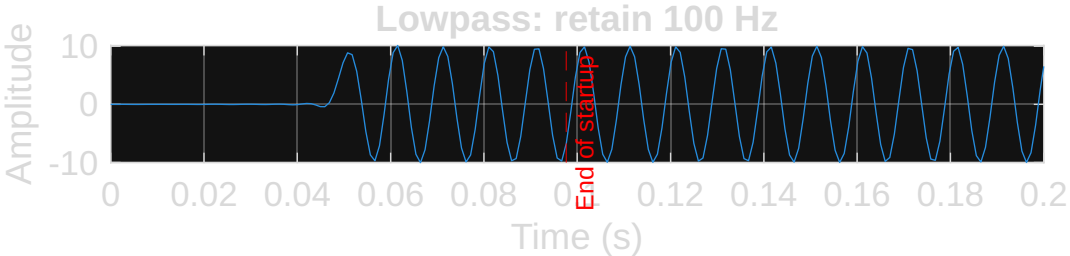


Bandpass: retain 200 Hz

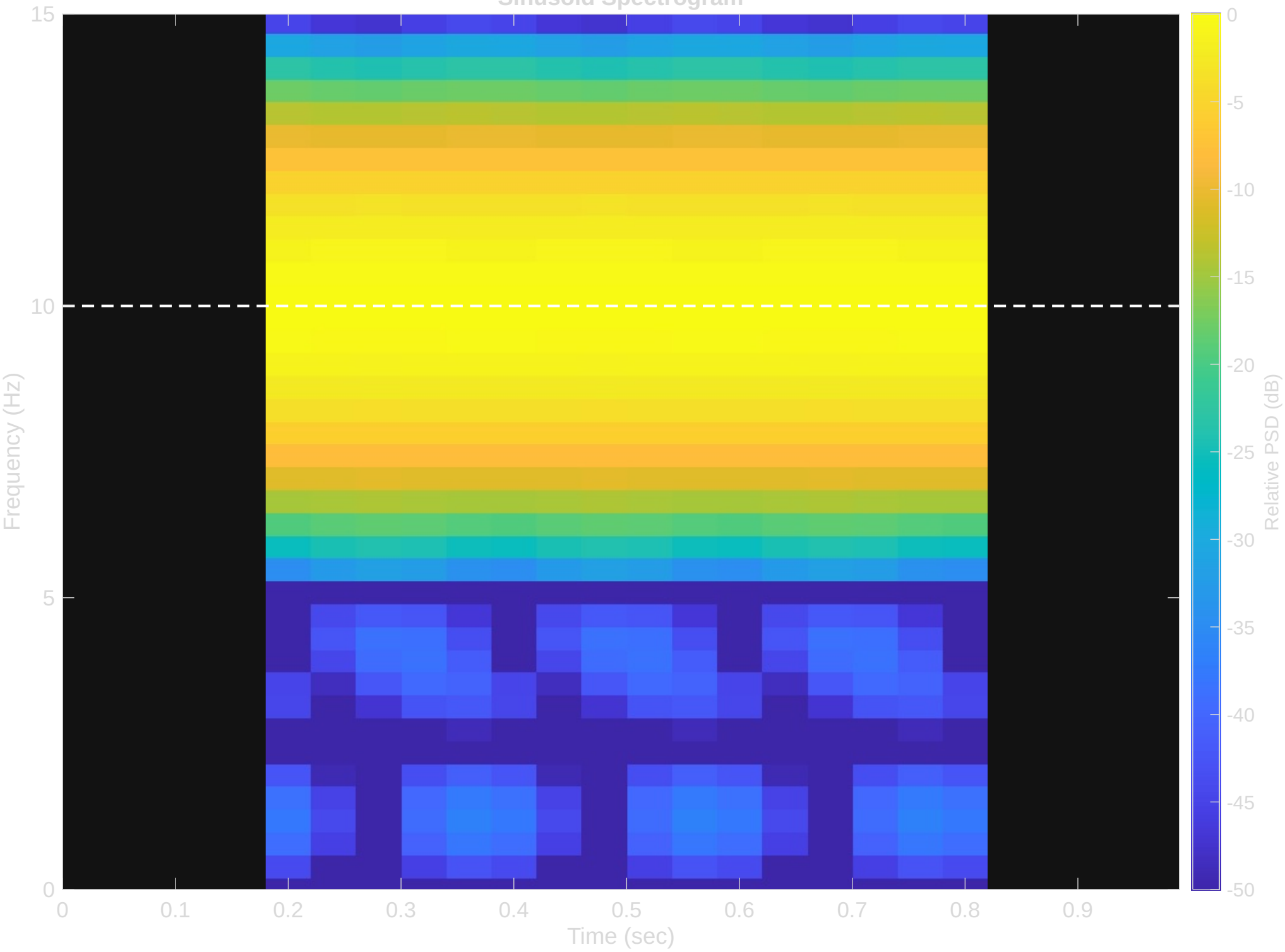


Highpass: retain 300 Hz

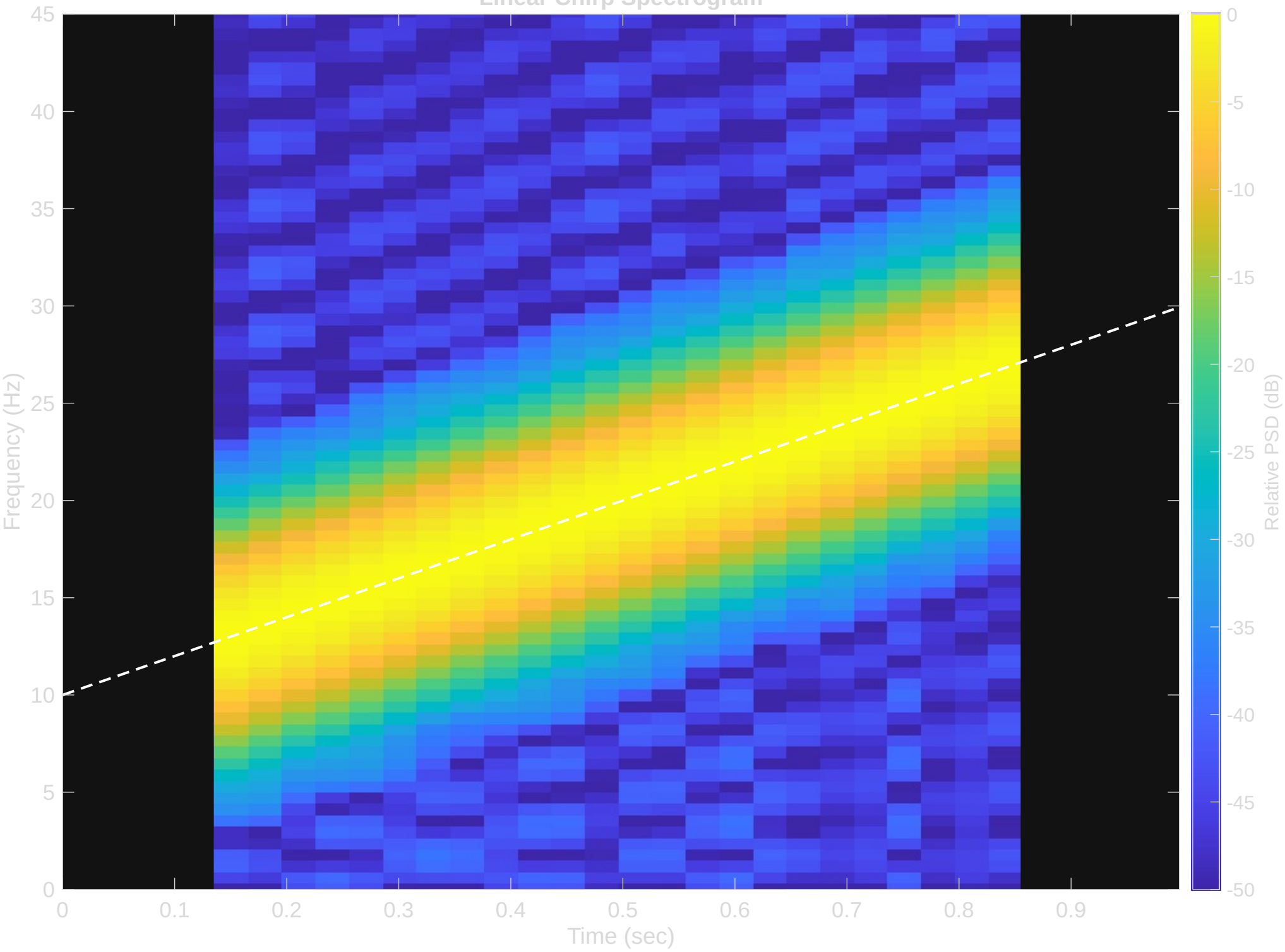




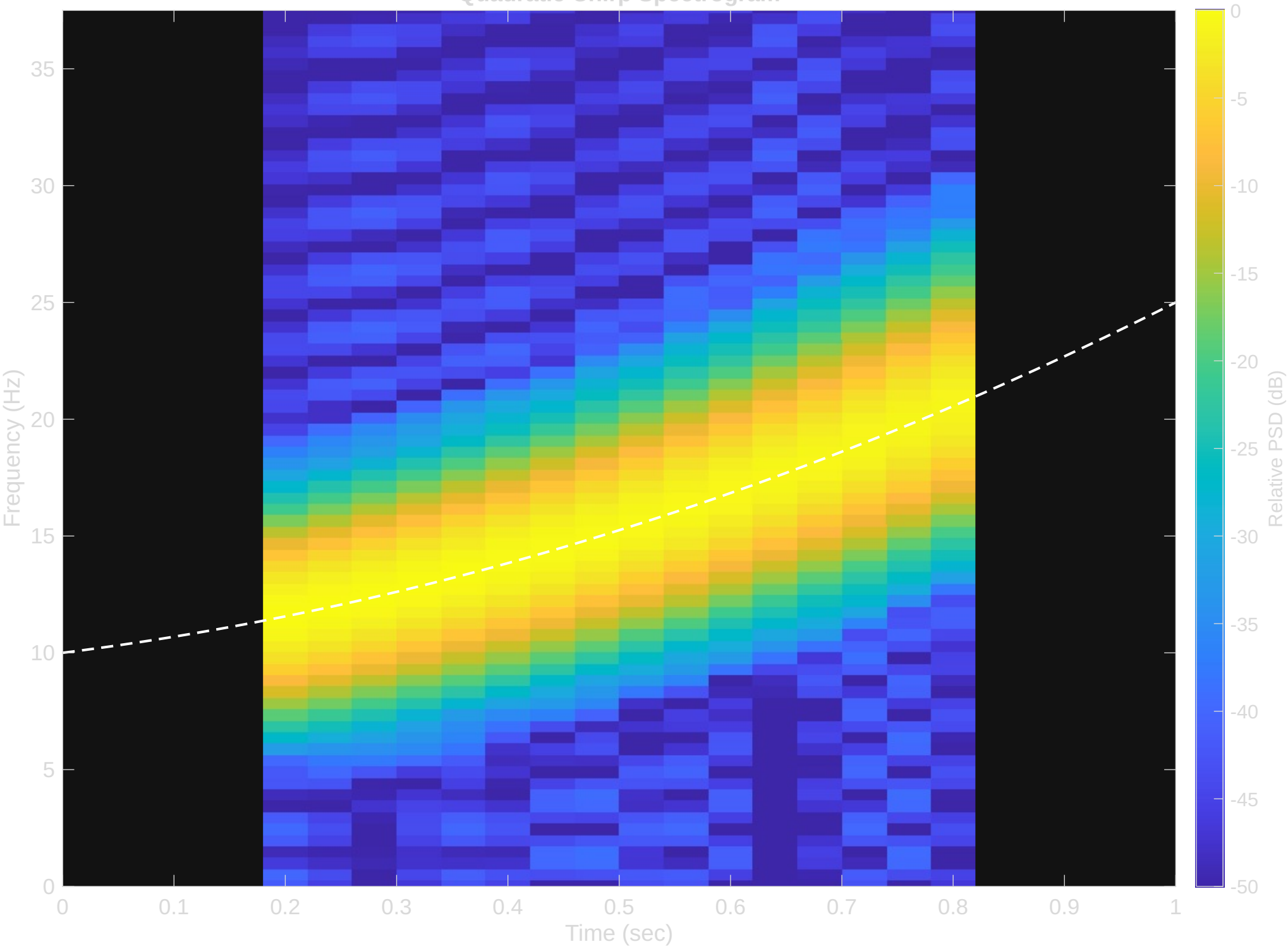
Sinusoid Spectrogram



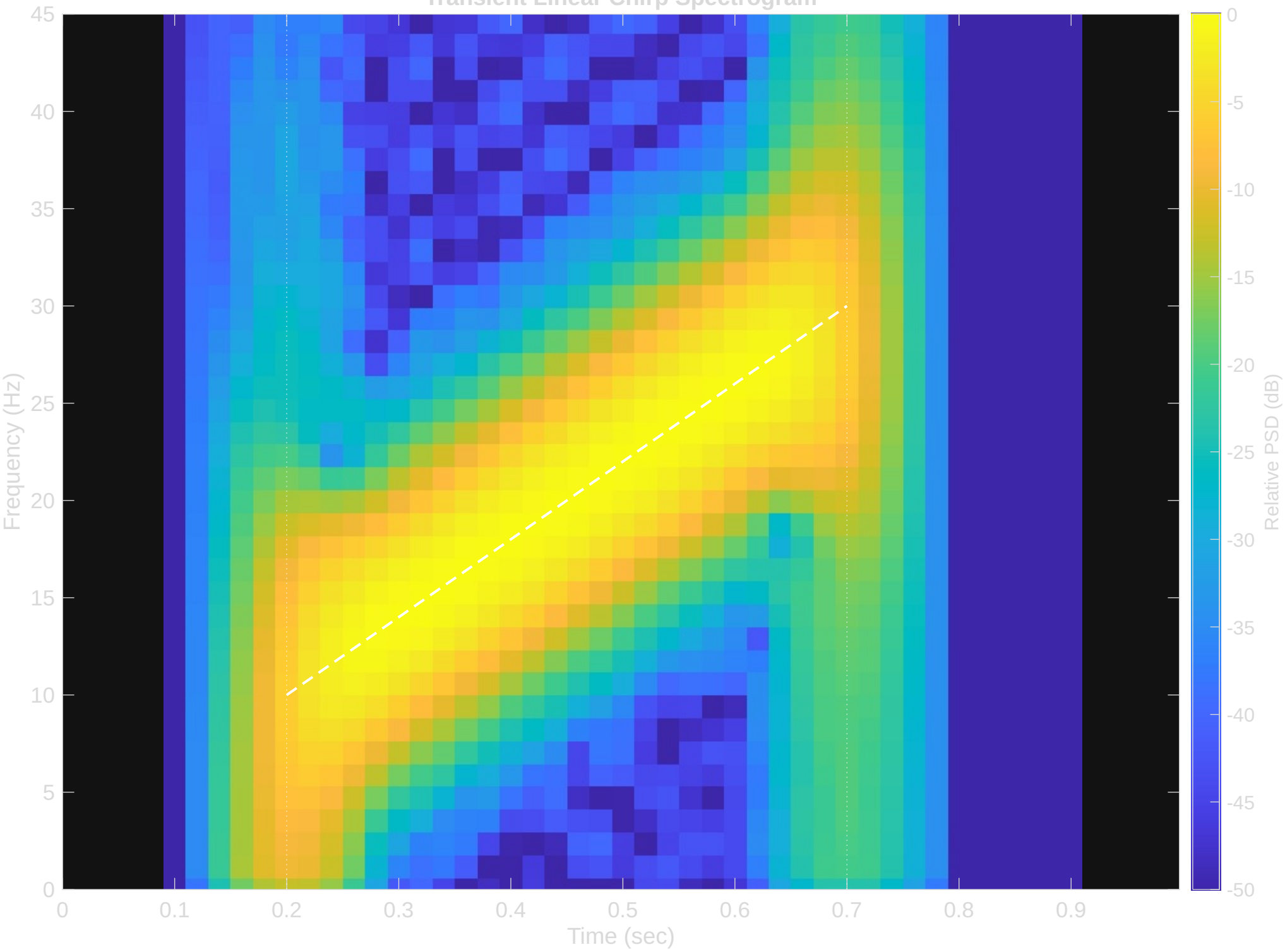
Linear Chirp Spectrogram



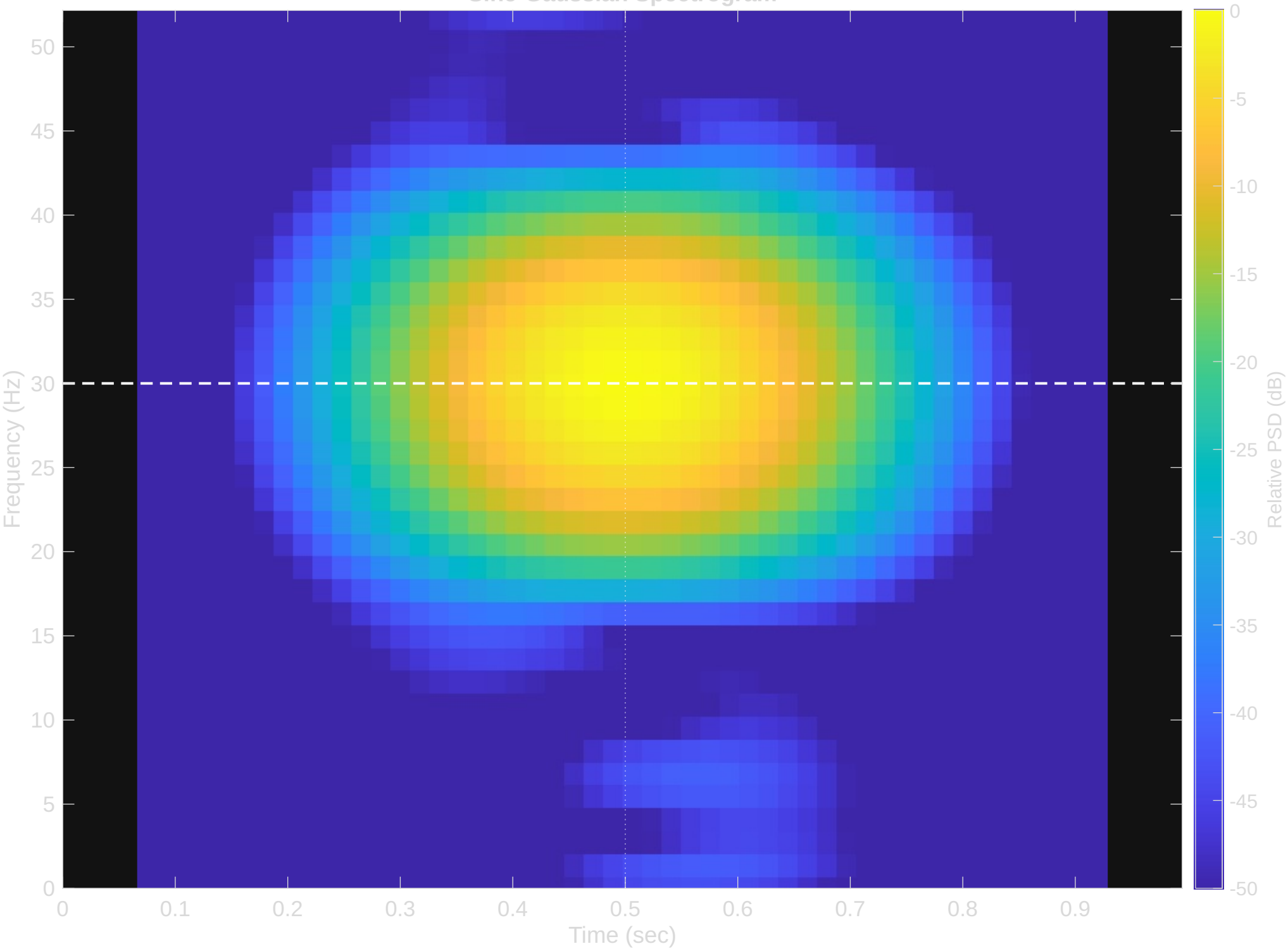
Quadratic Chirp Spectrogram



Transient Linear Chirp Spectrogram



Sine-Gaussian Spectrogram



Amplitude Modulation Spectrogram

